in countries with no tradition of maintaining large fleets, such as the United States and Germany. Thus a global naval arms race gathered momentum. America and Japan in particular embarked on major naval construction programs, followed by Germany. By 1914 the French had become the world's fifth-ranking naval power. The United States flexed the muscles of its new navy in a one-sided war with Spain in 1898, and then proudly sent the "Great White Fleet" around the world in 1907–09 in a pageant of American seapower. But it was the Russo-Japanese war of 1904–05 that provided the first thorough test of the new steam-and-steel navies. At the battle of Tsushima the Japanese were triumphant in a fleet encounter consciously inspired by Trafalgar, yet contested by ships capable of a speed of 16 knots with guns engaging effectively at a range of over 3 miles (5 km).



**Super-dreadnought**
The Royal Navy's *Queen Elizabeth* was an oil-fired batteship launched in 1913. With 15in guns and a top speed of 23 knots, she far surpassed the battleship *Dreadnought*, launched only seven years earlier.

## BIG SHIPS AND SNEAKY DEVICES

Early 20th-century navies were involved in an arms race that led to every ship launched being bigger and more heavily armed than the last. The intense rivalry between Britain and Germany, in particular, triggered an astounding burst of naval construction, which proved one of the main causes for Britain entering World War I on the side of France.

The race started in 1906 when the Royal Navy's HMS *Dreadnought* set a new standard for battleships. However, the *Dreadnought* was soon surpassed by the oil-fired super-dreadnoughts, such as *Conqueror*, while cruisers put on weight to become battlecruisers, faster ships with more powerful guns. These mighty ships attracted huge public and press interest possibly in part because of the great expense involved in their production. The main threats to these great ships—that naval commanders were well aware of—were mines and torpedoes. These were relatively cheap weapon systems against which ships had little protection.

When the long-awaited Great War came in 1914, the mighty fleets disappointed naval enthusiasts, avid for a repeat of the Battle of Trafalgar. Just as in the land war on the Western front, the new maritime technology favored defensive tactics. The threatening presence of submarines and mines militated against the bold, aggressive use of large warships. Even when sailing with a defensive screen of destroyers, battleships and battlecruisers proved vulnerable to these devices. The Royal Navy's Grand Fleet spent much of the war out of U-boat range, only occasionally risking a foray into the North Sea. While the one full-scale encounter between the British and German fleets at Jutland in 1916 was indecisive, commerce raiding, in the form of U-boat attacks on merchant shipping, almost crippled Britain's war effort. By the end of World War I, with experiments in flying aircraft off ships becoming more successful and techniques of submarine and anti-submarine warfare becoming ever more sophisticated, a new profile of naval war was emerging that was no longer dependent on ships engaging with another at close quarters.



**1889** Triple-expansion engines fitted in British battleships

**1894** Battle of Yalu River; Japan defeats China in First Sino-Japanese War (1894–95)

**1904** Japanese attack Russian fleet at Port Arthur

**1914** Start of World War I: the first aircraft carrier laid down

**1916** The battle of Jutland between Britain and Germany is the only battle fought between dreadnought battleships

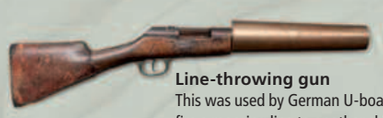
**1918** Germany surrenders following military collapse and mutiny in the navy

1880 | 1890 | 1900 | 1910 | 1920

**1880s** Following the example of Britain's Royal Navy, the use of rifled breech-loaders becomes widespread in modern navies

**1898** Admiral von Tirpitz begins expansion of German navy, triggering arms race with Britain

**1906** HMS *Dreadnought* is the first "all-big-gun" battleship with all-turbine propulsion



**Line-throwing gun** This was used by German U-boats to fire a mooring line to another ship.

**1915** Sinking of passenger ship SS *Lusitania* by German U-boat helps to bring the US into the war